



# UNIT – 4

# MAXIMA & MINIMA



## Method Of Finding Maxima & Minima (also called stationary or extreme or turning values) :

Given a function  $f(x, y)$

- i) Solve  $\frac{\partial f}{\partial x} = 0, \frac{\partial f}{\partial y} = 0$  simultaneously for  $x$  &  $y$ .
- ii) See that  $\frac{\partial^2 f}{\partial x^2} \cdot \frac{\partial^2 f}{\partial y^2} > \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2$  for the values of  $x, y$  obtained in (i)
- iii) When this condition is satisfied  $f(x, y)$  is maximum, if

$$\frac{\partial^2 f}{\partial x^2} \left( \text{or } \frac{\partial^2 f}{\partial y^2} \right) < 0 \text{ \& } f(x, y) \text{ is minimum if } \frac{\partial^2 f}{\partial x^2} \left( \text{or } \frac{\partial^2 f}{\partial y^2} \right) > 0$$

At  $x = x_1$  &  $y = y_1$  obtained in (i) let us denote

$$r = \frac{\partial^2 f}{\partial x^2}, \quad s = \frac{\partial^2 f}{\partial x \partial y}, \quad t = \frac{\partial^2 f}{\partial y^2} \quad \boxed{rt - s^2 > 0}$$

For maxima & minima we must have

$f(x, y)$  is maximum if  $r(\text{or } t) < 0$

$f(x, y)$  is minimum if  $r(\text{or } t) > 0$



1) Find the stationary value of  $x^3 + y^3 - 3axy$ ,  $a > 0$

Soln: We have  $f(x, y) = x^3 + y^3 - 3axy$

$$f_x = 3x^2 - 3ay, \quad f_y = 3y^2 - 3ax$$

$$r = f_{xx} = 6x, \quad s = f_{xy} = -3a, \quad t = f_{yy} = 6y$$

We now solve,  $f_x = 0$ ,  $f_y = 0$  as simultaneous equations.

$$\therefore x^2 - ay = 0 \quad \& \quad y^2 - ax = 0$$

To eliminate  $y$  we put  $y = \frac{x^2}{a}$  in the second equation.

$$\therefore x^4 - a^3x = 0 \quad \therefore x(x^3 - a^3) = 0$$

Hence  $x = 0$  or  $x = a$ .

When  $x = 0, y = 0$  & when  $x = a, y = a$ .

We therefore, have to examine the pairs

$$x = 0, y = 0 \quad \& \quad x = a, y = a.$$

i) For  $x = 0, y = 0$ ,

$$r = f_{xx} = 0, \quad s = f_{xy} = -3a \quad \& \quad t = f_{yy} = 0$$

Hence  $rt - s^2$  is -ve

$\therefore$  we reject this pair.

ii) For  $x = a, y = a$ ,

$$r = f_{xx} = 6a, \quad s = f_{xy} = -3a \quad \& \quad t = f_{yy} = 6a$$

$$\therefore rt - s^2 > 0$$

$\therefore f(x, y)$  is stationary (either maximum or minimum) at  $x = a, y = a$ .

But  $r = f_{xx} = 6a > 0 \therefore f(x, y)$  is minimum at  $x = a, y = a$

Putting  $x = a, y = a$  in  $x^3 + y^3 - 3axy$  the minimum value of  $f(x, y) = a^3 + a^3 - 3a^3 = -a^3$ .



Find the maximum & minimum values of  
 $x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$ .

Soln: We have  $f(x, y) = x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$

$$f_x = 3x^2 + 3y^2 - 6x, \quad f_y = 6xy - 6y$$

$$r = f_{xx} = 6x - 6, \quad s = f_{xy} = 6y, \quad t = f_{yy} = 6x - 6$$

We now solve,  $f_x = 0$ ,  $f_y = 0$  as simultaneous equations.

$$\therefore x^2 + y^2 - 2x = 0 \quad \& \quad xy - y = 0 \text{ i.e. } y(x - 1) = 0$$

$$\therefore y = 0 \text{ or } x = 1.$$

$$\text{When } y = 0, \quad x^2 - 2x = 0 \quad \therefore x = 2 \text{ or } x = 0$$

$$\& \text{ when } x = 1, \quad y^2 - 1 = 0 \quad \therefore y = 1 \text{ or } y = -1$$

i) For  $x = 0, y = 0$ ,

$$r = f_{xx} = -6, \quad s = f_{xy} = 0 \quad \& \quad t = f_{yy} = -6$$

$$\therefore rt - s^2 > 0$$

$f(x, y)$  is stationary at  $x = 0, y = 0$ .

But  $r = f_{xx} = -6 < 0 \quad \therefore f(x, y)$  is maximum at  $x = 0, y = 0$  & the maximum value is 4.

ii) For  $x = 2, y = 0$ ,

$$r = f_{xx} = 6, \quad s = f_{xy} = 0 \quad \& \quad t = f_{yy} = 6$$

$$\therefore rt - s^2 > 0$$

$f(x, y)$  is stationary at  $x = 2, y = 0$

But  $r = f_{xx} = 6 > 0 \quad \therefore f(x, y)$  is minimum at  $x = 2, y = 0$  & the minimum value is 0.

iii) At  $x = 1, y = 1$  or at  $x = 1, y = -1$

$$rt - s^2 \text{ is -ve}$$

$\therefore$  we reject this pair.



- 3) Find all the stationary values of  $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$ .
- 4) Show that minimum value of  $u = xy + a^3 \left( \frac{1}{x} + \frac{1}{y} \right)$  is  $3a^2$ .